

$$\therefore |a - ab_mean| = |K_1 - K_2| = |-(K_2 - K_1)| = |K_2 - K_1| = |b - ab_mean|$$

$\therefore$ ~~the~~ $c = ab_mean$ is a valid solution. $\square$ (Existence part)

Uniqueness: Let, there exist two integers c_1 and c_2 s.t.

$$|a - c_1| = |b - c_1| \text{ and } |a - c_2| = |b - c_2|$$

It's easy to see that for both eqns, ~~both~~ the differences cannot be both positive and negative together, otherwise we will have $a = b$, which is a contradiction.

Case 1: $a - c_1 > 0, b - c_1 < 0, a - c_2 > 0, b - c_2 < 0$

$$\therefore |a - c_1| = a - c_1 = |b - c_1| = -(b - c_1)$$

$$\Rightarrow 2c_1 = (a+b) \Rightarrow c_1 = \left(\frac{a+b}{2}\right)$$

$$\therefore |a - c_2| = a - c_2 = |b - c_2| = -(b - c_2)$$

$$\Rightarrow 2c_2 = (a+b) \Rightarrow c_2 = \left(\frac{a+b}{2}\right). \therefore c_1 = c_2$$

Case 2: $a - c_1 > 0, b - c_1 < 0, a - c_2 < 0, b - c_2 > 0$

$$\therefore |a - c_1| = a - c_1 \quad |b - c_1| = -(b - c_1)$$

$$a - c_1 = -(b - c_1) \Rightarrow 2c_1 = (a+b) \Rightarrow c_1 = (a+b)/2$$

$$|a - c_2| = |b - c_2| \Rightarrow -(a - c_2) = b - c_2 \Rightarrow 2c_2 = (b+a) = (a+b) \Rightarrow c_2 = (a+b)/2$$

$$\therefore c_1 = c_2$$

Case 3: $a - c_1 < 0, b - c_1 > 0, a - c_2 > 0, b - c_2 < 0$

This case is similar to Case 2.

Case 4: $a - c_1 < 0, b - c_1 > 0, a - c_2 < 0, b - c_2 > 0$

This case is similar to case 1.

$$\therefore c_1 = c_2 \quad \square$$